Comparative Study of Post-Hoc Methods vs Intrinsic Interpretability in Vision Models

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The "Black Box" Problem

- Deep learning models (CNNs, ViTs) have opaque decision-making processes.
- **Post-Hoc XAI**: Explainers applied *after* training (e.g., **SHAP** [?], **LIME** [?]).
- **Intrinsic XAI**: Models interpretable by design (e.g., **ProtoPNet** [?]).

Mathematical Foundations: XAI Methods

SHAP (Shapley Values)

$$\phi_i = \sum_S \frac{|S|!(n - |S| - 1)!}{n!} \Delta v$$

GradCAM

$$L^c_{Grad-CAM} = ReLU\left(\sum_k \alpha^c_k A^k\right)$$

LIME (Local Surrogate)

$$\xi(x) = \arg\min_{g \in G} \mathcal{L} + \Omega(g)$$

ProtoPNet Similarity

$$sim = \log\left(\frac{d^2 + 1}{d^2 + \epsilon}\right)$$

Mathematical Foundations: Metrics

Infidelity

$$INFD = \mathbb{E}_I\left[\left(f(x) - f(x - I) - I^T\Phi(x)\right)^2\right]$$

Complexity (Entropy)

$$C(\Phi) = -\sum_i p_i \ln(p_i)$$

Localization (IoU)

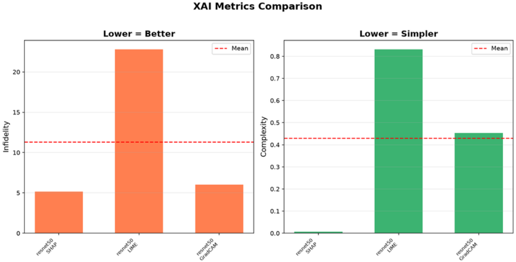
$$Loc = \frac{|M \cap B|}{|M \cup B|}$$

Rank Agreement (Spearman)

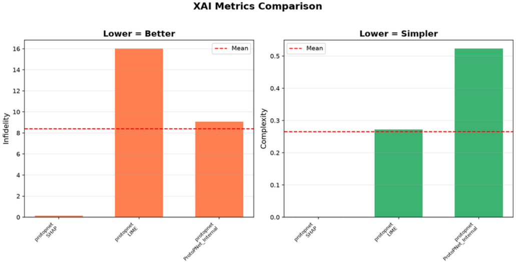
$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2 - 1)}$$

Experimental Setup & Metrics

Post-Hoc Metrics



Intrinsic Metrics

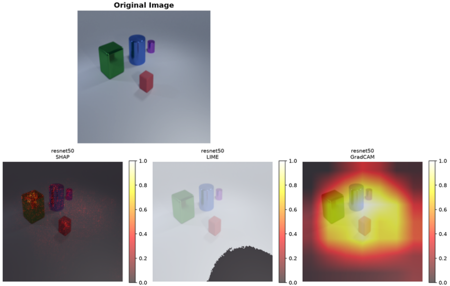


Datasets: *CLE4EVR* (3D objects) and *MNMath* (Handwritten equations).

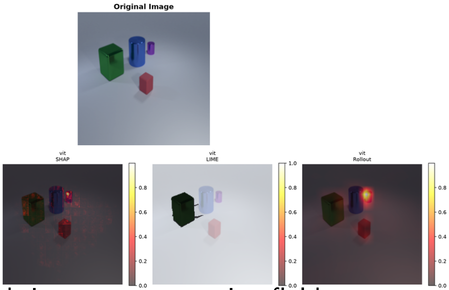
Key Metrics: *Infidelity*, *Complexity*, *Localization*.

Post-Hoc Methods on CLE4EVR Dataset

ResNet50



Vision Transformer (ViT)



- **GradCAM** performs well: large 3D objects match its coarse receptive field.
- **LIME** underperforms: default SLIC superpixels severely fragment the smooth 3D shapes.

LIME Ablation Study on CLE4EVR

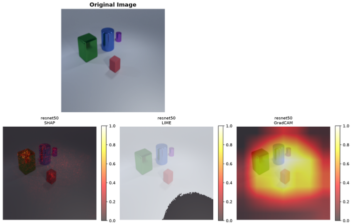
Hypothesis: Does matching LIME's superpixel count to the number of objects improve explanations?

- **Default ($K = 50$)**: Fragments smooth 3D shapes.
- **Matched ($K \approx 4$)**: Visually aligns with boundaries, but *Infidelity doubles* (from 16.0 to 34.2).

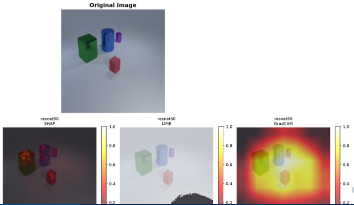
Conclusion: Linear surrogates fail to capture the non-linear reasoning of CNNs.

Superpixels (K)	Alignment	Infidelity $\downarrow$
Default (50)	Fragmented	16.010
Match (4)	Object-Aligned	34.250

Default LIME ($K = 50$)

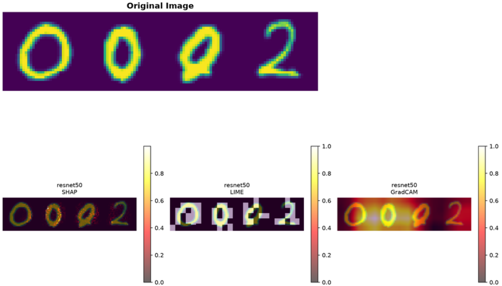


Ablated LIME ($K \approx 4$)

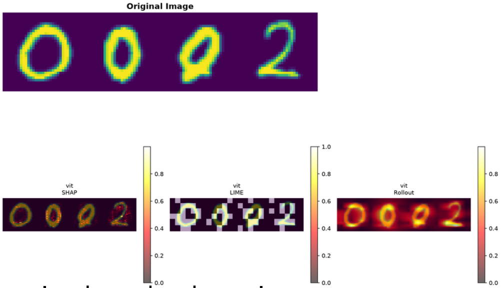


Post-Hoc Methods on MNMath Dataset

ResNet50

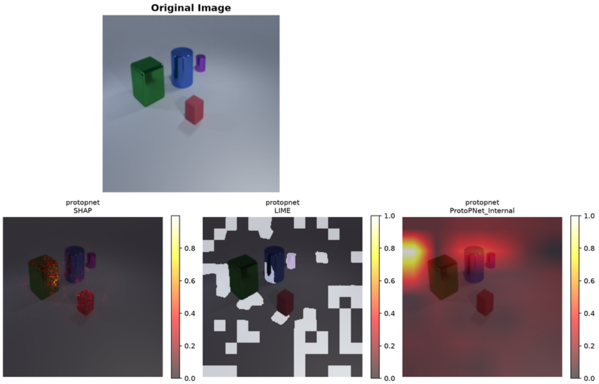


Vision Transformer (ViT)



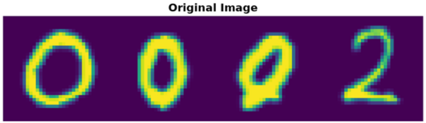
- **SHAP** heavily outperforms GradCAM on fine-grained stroke detection.
- GradCAM's low spatial resolution blurs over the thin handwritten digits.

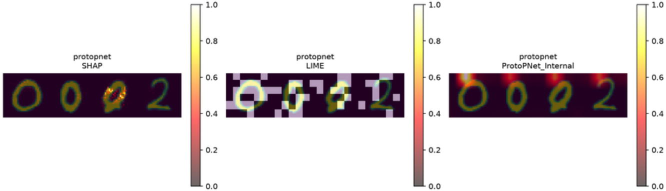
Visual Comparison on CLE4EVR



SHAP isolates edges (sparse), while **ProtoPNet Internal** highlights broad, context-aware regions.

Visual Comparison on MNMath



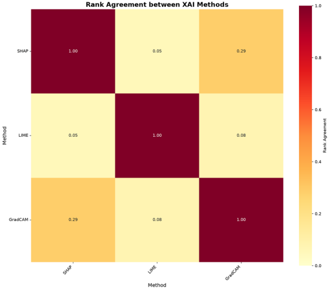


ProtoPNet reasoning encompasses the whole digit and its context, completely ignoring the strict "stroke" pixels favored by SHAP.

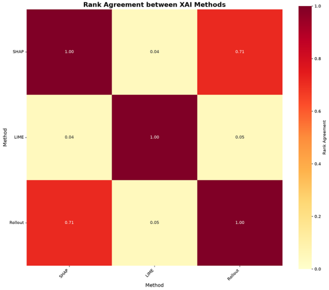
Rank Agreement: CLE4EVR Dataset

- Correlation between Spatial Locality (Intersection) and Infidelity.
- Across all architectures, post-hoc rank correlation is **near zero or negative**.

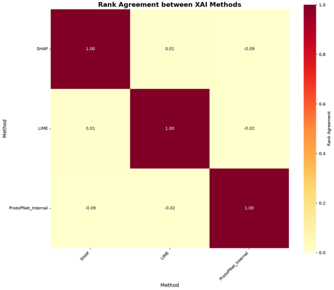
ResNet50



ViT



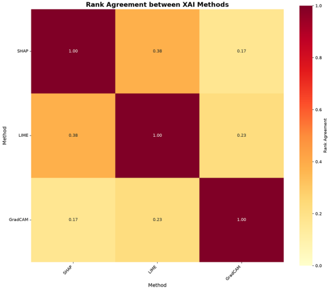
ProtoPNet



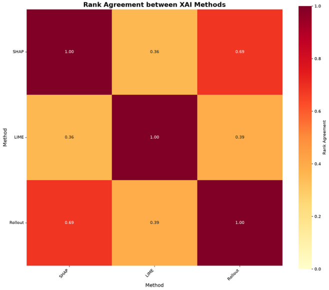
Rank Agreement: MNMath Dataset

- The disconnect is even worse on fine-grained data (-0.24 for ProtoPNet).
- Post-hoc methods misrepresent the actual broad spatial reasoning of the network!

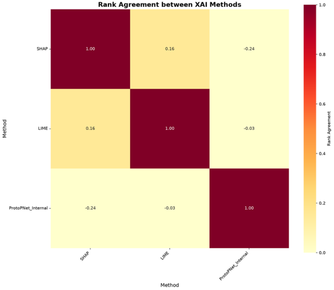
ResNet50



ViT



ProtoPNet



Conclusion

Key Takeaways

- Relying strictly on "Localization Accuracy" is dangerous: it assumes networks reason using human-defined object boundaries.
- Intrinsic interpretability reveals that models learn broad spatial contexts, unlike what post-hoc methods suggest [**?**].

Thank You!

Questions?

References I